

Utilization of Fly Ash and Rice Husk Ash in Cold Mix Asphalt as Filler

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Abstract. Road infrastructures are pivotal to the development of infrastructure, transport and overall economy of any country which has a detrimental effect on the environment. So, adopting an eco-friendly pavement technology with the utilization of waste is of paramount importance. In this study, Cold mix asphalt (CMA) is produced with fillers like Fly ash (FA) and Rice husk ash (RHA) and parameters like Marshall stability, Marshall quotient, Indirect tensile strength, moisture susceptibility and rutting resistance of CMA mixes were determined through laboratory tests. Replacing SD with RHA and FA can improve the Marshall stability to 20.32% and 25.99% respectively. The Marshall quotient value of mixes with SD, RHA and FA are 1.93 kN/mm, 2.68 kN/mm and 3.01 kN/mm at 8%, 9% and 8% optimum binder content respectively. The betterment in tensile strength and moisture susceptibility of mixes was also found. The rutting deformation of the mix with SD is 16.57 mm, the highest among other filler mixes and the resistance has increased to 1.66 % and 52.16% after the application of RHA and FA filler. Finally, a detailed cost analysis was carried out and came upon that the mix with FA is the most cost-effective mixture.

1 Introduction

Cold mix asphalt (CMA) comprising of bitumen emulsion, aggregates, fillers and water [1]. It is an eco-friendly technology with significant advantages like less fuel consumption, no heating of binder and aggregate, an easy mixing process etc. Besides these benefits, it has some major drawbacks like lower strength at the initial stage, high moisture damage and high air voids [2,3,4]. Different researchers have utilized several fillers, waste materials, and additives in CMA to eliminate these issues.

The fillers are the finest portion of aggregate that occupies the space within the mixture and are added upto 12% by weight of mixes [5]. The filler stiffens the binder [6], satisfying the gradation and volumetric requirements of the asphalt mix [7]. Different waste materials were utilized in CMA as fillers to enhance the properties of the mixture. Some

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developing countries like India have produced 960 million solid wastes, usually managed by dumping into landfills and incineration. These wastes need to be managed or recycled by utilizing those as a component of the mix to ensure safe disposal. Rice husk ash (RHA) and Fly ash (FA) are the most commonly generated industrial wastes globally and adding these materials as filler in CMA can increase its field utilization as well as it has reduced the landfills.

Fly ash is the by-product material manufactured from flue ash's mechanical or electrostatic precipitation during coal combustion of power stations [8]. Fly ash is widely used in soil stabilization, concrete industry, and filler material in the asphalt mix. It has the pozzolanic property, which provides a cementitious effect when combined with water. It is preferable to use in cold mixes. Using FA can improve cold mixes' stability to 150% [9]. The use of FA as filler at 0-6% in CMA can enhance the performance similar to hot mix asphalt. The use of FA can also improve the stiffness of CMA [10].

The use of RHA is also beneficial when it is utilized in the asphalt mix. RHA is mainly produced by burning the rice husk in an uncontrolled process that is cheaper and convenient but involves environmental pollution. In many literatures, the partial or full replacement of conventional filler with RHA was observed and reported superior performance compared to the conventional mix. The substitute of 50% limestone filler with RHA has bettered the Marshall stability and volumetric properties of the asphalt mix [11]. The particles of RHA reinforce the binder and subsequently improve the stiffness and cohesion, which automatically enhance the Marshall stability of the mix [12, 13]. The asphalt mixes modified with RHA had more significant resistance to moisture cracking and deformation and long-term aging compared to Ordinary Portland cement and hydrated lime-modified mixes [12, 13]. However, the porous structure of RHA particles detrimentally affects the performance like stability, moisture susceptibility, fatigue and rutting of mixes than those with stone dust [14].

These fillers are primarily utilized in HMA for improving the performance. However, there is a need to utilize these fillers in CMA which was not extensively explored by existing works of literature. In this experimental study, FA and RHA are added at 5% by weight of the mixture in CMA as filler by replacing conventional stone dust.

2 Materials

The cationic bitumen emulsion is utilized as a binder in CMA. The bitumen emulsion was collected from Bitchem Asphalt Technologies Limited, Guwahati. The specific gravity of bitumen emulsion is 1 and storage stability of 0.1%. The penetration and ductility of bitumen emulsion on its residue are 82 mm and 87 cm respectively. The river stone aggregate was utilized in this study which has the maximum and minimum nominal size of 13.2 and 0.075 mm respectively. The continuously graded aggregate gradation is as per IRC SP:100:2014 [15] (suggested for the surface course), having the specific gravity and water absorption value of 2.62 and 1.1%. The elongation and flakiness index of aggregate is 16.8% and 10.5% respectively. The impact and abrasion value of aggregate is 23.1% and 28.65%. The RHA and FA fillers passing through 0.075 mm sieve were incorporated in CMA. The maximum amount of filler is added at 5% by mixture weight. The specific gravity of RHA and FA is 2.05 and 2.07, respectively.

3 Experimental Programme

3.1 Coating test

The coating test was conducted to finalize the water dosage required to properly mix CMA. The test displays the degree of affinity and compatibility between the mineral aggregate and bitumen emulsion. The test is conducted at room temperature. The starting point of water is 2% by weight of aggregate with an increment of 1%. The fixed amount of bitumen emulsion is added and mixed until an evenly coated mixture is found [16]. The pre-wetting water appears to be a viscosity-reducing agent that promotes the workability and helps to coat the aggregate surfaces [16].

3.2 Marshall stability test

The Marshall test is a mix design procedure, widely used for CMA in most countries. The Marshall stability test determines the optimum binder content (OBC) as per ASTM D6927-15 [17]. The volumetric properties of mixes are also calculated from the test results. Marshall test is used to quantify the ability to counteract the plastic deformation of cylindrical CMA samples under gradual loading of 50 mm/min along with the cylindrical body [17]. The flow and stability value also calculates Marshall quotient.

3.3 Indirect tensile strength and moisture susceptibility test

The tensile strength of prepared Marshall samples (dry conditioned) were evaluated as per ASTM D6931-17 [18]. The tensile strength of mixes is used to enumerate the quality of the asphalt mix in conjunction with the potential of rutting and fatigue failure. The samples were taken and performed ITS test after a specific curing period. Cylindrical Marshall samples are subjected to the diametrical loading rate of 50 ± 5 mm/min. The peak load at failure is determined as the tensile strength of the mixes. The ITS for each mix was calculated by:

$$ITS = \frac{2P_{\max}}{\pi td}$$

Where, $P_{\max}$ = Load (kN), t = Samples height and d = diameter of the sample.

Moisture susceptibility is a major mix design criteria that indicates the reduction in tensile strength in the presence of water. The consequences of environmental changes along the service life of pavement is measured in the laboratory by Tensile strength ratio (TSR) to find the durability of an asphalt mix. AASHTO T283 is adopted to encounter the moisture damage of CMA and HMA [19]. The TSR of the mixes is calculated by:

$$TSR = \frac{ITS \text{ of conditioned sample}}{ITS \text{ of unconditioned sample}} \times 100$$

3.4 Rutting test

Rutting failure is the longitudinal depression along the path of the wheel. This phenomenon is due to the consolidation of subgrade soil caused by repeated traffic loading, lateral movement of pavement materials, shear failure of asphalt pavement or a combination. The permanent deformation of CMA with FA and RHA fillers was determined by undergoing

150 mm and 63.5 mm dia specimens under repeated wheel passes of 50 nos/minutes in the Hamburg wheel tracking device [20]. The specimen should have an equal density to the Marshall specimen. The wheel can provide a load of 705 ± 4.5 N on the rutting specimens. The test temperature is 60°C [21] and the frequency of 50 nos of wheel passes per minute with the boundary condition of 10,000 cycles [22].

4 Results and discussion

4.1 Marshall stability and volumetric properties of CMA

The optimum premixing water is 3% and 4% for the CMA with RHA and FA filler which was determined by varying the water dosage from 2-5% by weight of mixture and observing the coating ability of aggregate particles. The bitumen emulsion is added at 7-10% by weight of the mix and the OBC was determined at 8%, 9% and 8% for SD, RHA and FA respectively. The results are given in Fig. 1. The Marshall stability value of CMA with SD, RHA and FA is 10.04 kN, 12.08 kN and 12.65 kN. The improvement is observed in the Marshall stability value of CMA, results due to the hydration effect of FA and pozzolanic properties of RHA in the presence of existing water that stiffens the mixture. This phenomenon also provides a secondary bond in the mix [23]. The volumetric properties of mixes are given in Table 1.

Table 1. Volumetric properties of mixes

Name of filler	Bulk density (gm/cm ³)	Volume of voids in % (V _v)	Voids in mineral aggregate in % (VMA)	Voids filled with bitumen in % (VFB)
SD	1.98	9.16	24.52	62.65
RHA	2.11	7.91	30.09	73.71
FA	2.21	5.24	24.89	78.95

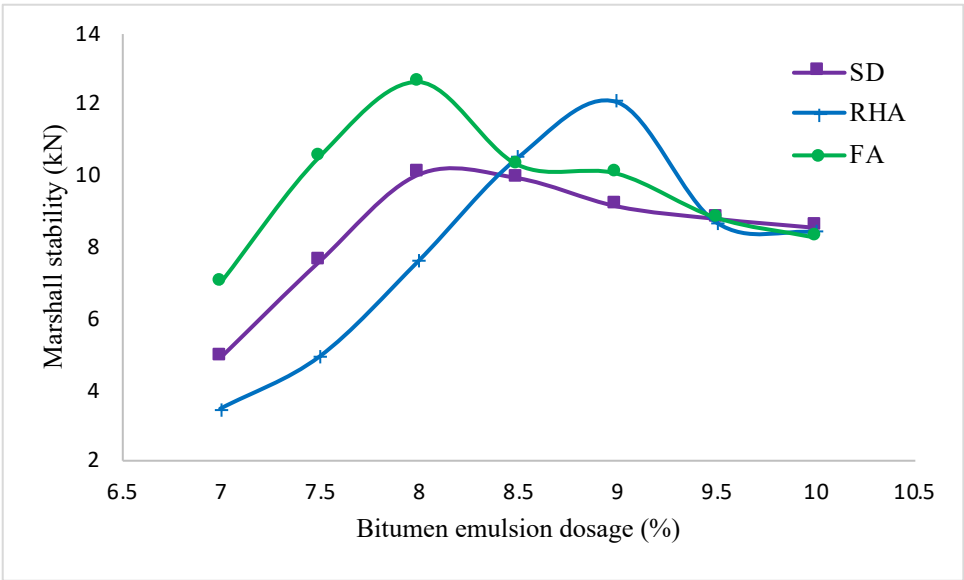


Figure 1. Marshall stability of CMA at different dosage of bitumen emulsion

1.2 Marshall quotient of CMA mixes

The Marshall quotient (MQ) value indicates the stiffness of the mix that is measured by dividing the stability value with flow value. A higher MQ value refers to the higher stiffness of the mix which can efficiently distribute the applied load and withstand the deformation caused by creep [24]. However, a very high MQ value means a brittle mix that results in bleeding and cracking. MORT&H has recommended the value range from 2-5 kN/mm for asphalt mix. The MQ value of mixes with different fillers is given in Fig. 2. The MQ value for the mix with SD filler is 1.93 kN/mm (<), indicating a less stiff mixture. Furthermore, the MQ of the mix with RHA and FA is 2.68 kN/mm and 3.01 kN/mm which refers to a robust mixture. The mixture with FA shows the highest MQ value as FA particles utilize the excess water for hydration [23].

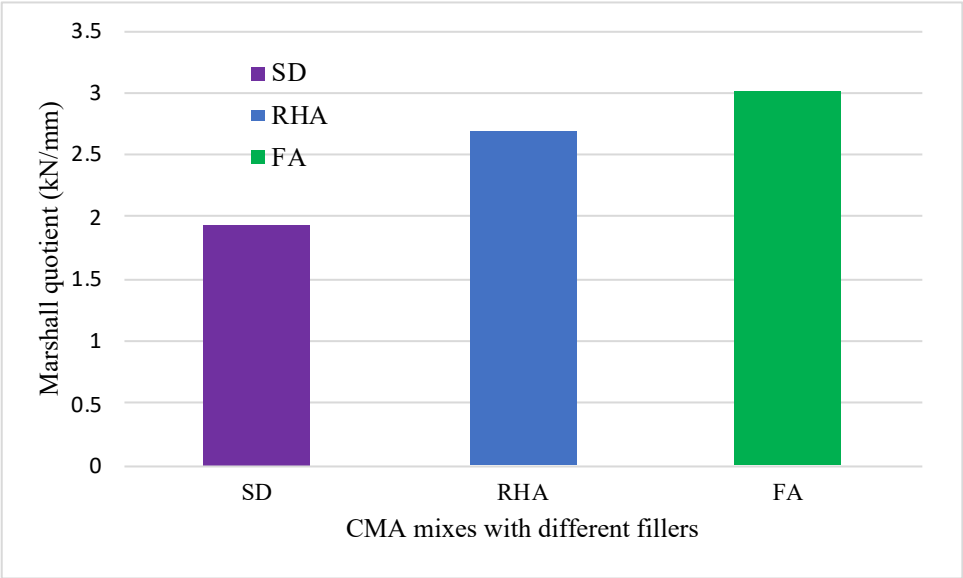


Figure 2 Marshall quotient values of CMA with different fillers

1.3 Indirect tensile strength (ITS) and tensile strength ratio (TSR) of CMA mixes

The ITS and TSR value of CMA with conventional SD filler is 200 kPa and 71.22% respectively. The minimum requirement of TSR value for asphalt mix is 80% as per AASTHO and 75% as per IRC SP:100:2014. ITS and TSR of mixes with RHA and FA are given in Fig. 3 and Fig. 4. It is observed that with the increasing dosage of bitumen emulsion, the ITS has increased and at OBC, the ITS value is highest following the lower ITS value. The trend is similar to the Marshall stability graph for both mixes. The ITS and TSR value of CMA with RHA filler is increased to 188.75% and 51.74% respectively, at 9% OBC. Similarly, the ITS and TSR of CMA with FA filler is increased to 33.69% and 17.38% separately. It is stated that the TSR value of CMA mixes exceeds the code requirements, which is satisfactory. The high resistance of mixes against moisture is due to the film of bitumen emulsion on the aggregate particles and the pozzolanic effect of fillers within the mixture.

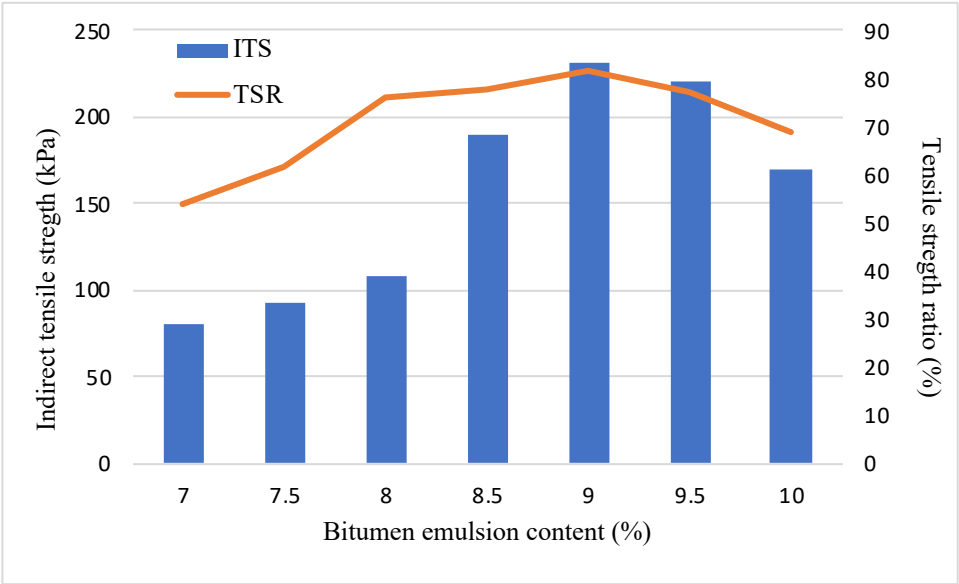


Figure 3 ITS and TSR of CMA with RHA filler

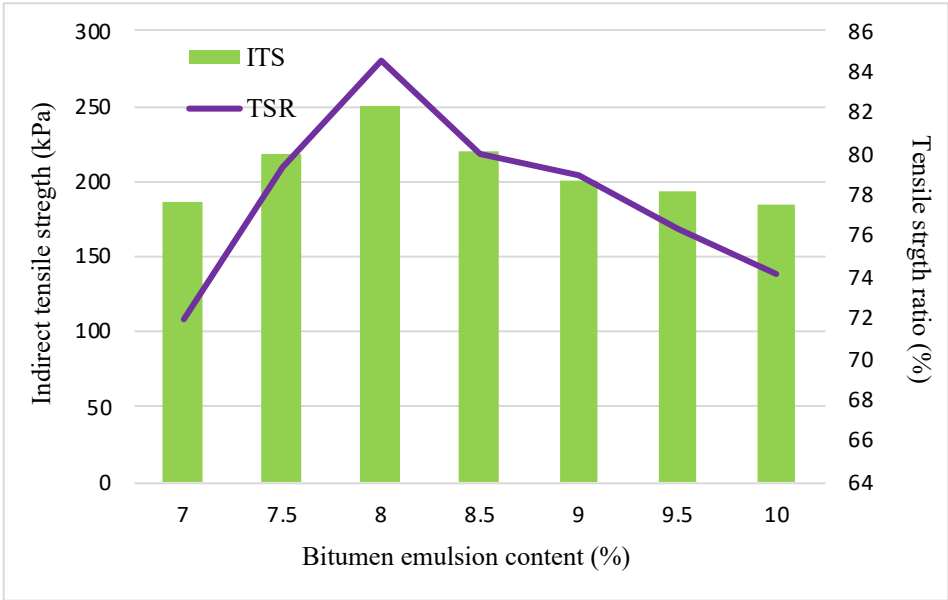


Figure 4 ITS and moisture susceptibility of CMA with FA filler

1.4 Rutting resistance of CMA mixes

The rutting test is an essential aspect in terms of determining the premature failure of the asphalt mix. The rutting samples were tested with the same boundary condition to compare the performance of the mixes. The test results are given in Figure 5. The rut depth is increasing with the increasing no. of wheel passes and the final rut depth was recorded at 10,000 wheel passes. It was noticed that the rut depth obtained for mix with SD filler is 16.57 mm which is the highest among other mixes. The rut depth achieved for CMA with RHA and FA is 16.3 mm and 10.89 mm respectively. The mixture with FA is the most rut resistant

under this test. The higher rut resistance of RHA and FA mixture is due to the pozzolanic behaviour of these fillers and the hydration mechanism of fillers particles within the mixture.

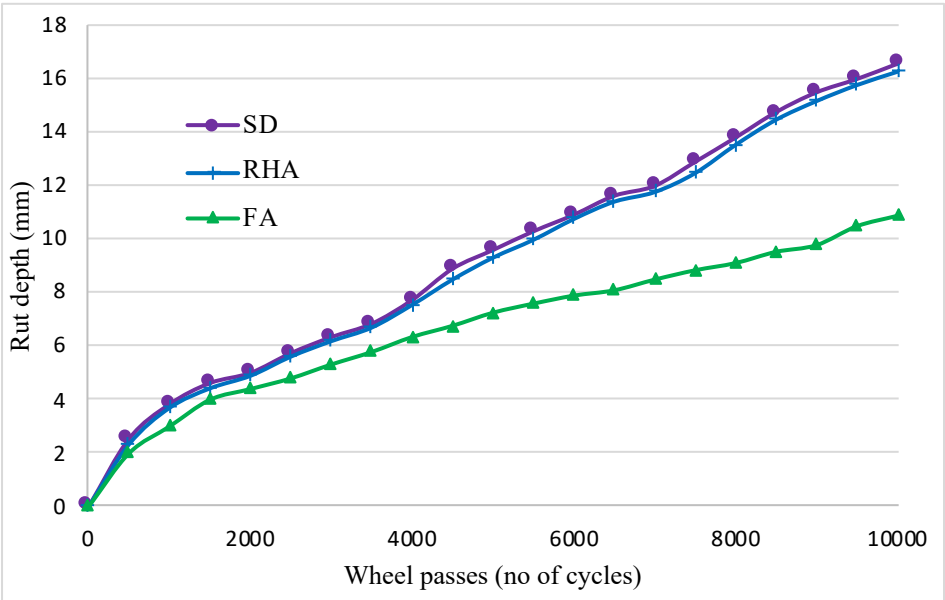


Figure 5 Rutting resistance of CMA with different fillers

1.5 Cost analysis

The utilization of waste materials in pavement raised another concern related to the cost-effectiveness of any technology. The cost of pavement with these fillers is calculated for a single lane 1 km road of 40 mm thick. The material and construction costs were calculated, including the cost of machinery and equipment. Following are the calculation for the required quantity of materials for single lane 1 km road:

- i. Quantity of material for CMA with SD filler
 - Density of mix with 5% SD filler and 8% bitumen emulsion content is 1980 kg/m³
 - Quantity of mix for 40 mm thick pavement layer $1000 \times 3.75 \times 0.04 \times 1980 = 297000$ kg $\cong$ 297 tons
 - Amount of bitumen emulsion required= 8% $\times$ 297=23.76 tons
 - Total quantity of aggregate with filler = (297-23.76)/1.5 = 182.16 tons
 - Amount of stone dust = 182.16 $\times$ 5%=9.108 tons
 - Quantity of aggregate = 182.16-9.108 = 173.052 tons
- ii. Quantity of material for CMA with FA filler
 - Density of mix with 5% FA filler and 8% bitumen emulsion content 2020 kg/m³
 - Quantity of mix for 40 mm thick pavement layer $1000 \times 3.75 \times 0.04 \times 2020 = 303000$ kg $\cong$ 303 tons
 - Amount of bitumen emulsion required= 8% $\times$ 303 =24.24 tons

- Total quantity of aggregate with filler = $(303 - 24.24)/1.5 = 185.84$ tons
- Amount of RCAD = $185.84 \times 5\% = 9.29$ tons
- Quantity of aggregate = $185.84 - 9.29 = 176.55$ tons

iii. Quantity of material for CMA with RHA filler

- Density of mix with 5% RHA filler and 9% bitumen emulsion content 2030 kg/m³
- Quantity of mix for 40 mm thick pavement layer $1000 \times 3.75 \times 0.04 \times 2030 = 304500$ kg $\cong 304.5$ tons
- Amount of bitumen emulsion required = $9\% \times 304.5 = 27.41$ tons
- Total quantity of aggregate with filler = $(304.5 - 27.41)/1.5 = 184.73$ tons
- Amount of RCAD = $184.73 \times 5\% = 9.24$ tons
- Quantity of aggregate = $184.73 - 9.24 = 175.49$ tons

The calculation of materials is given in Table 2 and the materials costs are also included. It is observed that the cost per ton of mix with FA is Rs. 7196.55/- per ton which is the lowest. However, the cost is higher for 1 km road length made with FA filler. This phenomenon is due to the higher density of the mix, which requires more materials. Additionally, the cost of CMA with RHA is Rs. 2412801.14 per lane km, i.e costliest mixture among them. As per the study, it is suggested that the best suitable filler for CMA is FA considering the performance and cost of the mixture.

Table 2 Cost calculation of mixes with different fillers

Sl. no.	Items	SD	FA	RHA	Remarks
1.	Bulk Density (kg/m ³)	1980	2020	2030	-
2.	Quantity of mix (tons/lane km)	297	303	304.5	-
3.	Quantity of bitumen emulsion (tons/lane km)	23.76	24.24	27.41	-
4.	Cost of bitumen	50500	50500	50500	Rate of Meghalaya Bitchem Pvt.Ltd.Assam,

	emulsion (Rs/ton)				(Including 18% GST)
5.	Filler cost (Rs/ton)	1100	1000 (Rate as per Skyward Concricks private Limited, Kolkata, West Bengal)	2000 (Rate as per Krishna Udyog, Kolkata, West Bengal)	
6.	Per ton cost of CMA with filler (Rs)	7207.73	7196.55	7923.81	Including, contractor profit, water charges, Chess and GST

5 Conclusions

The laboratory analysis compares the performance of the CMA mixture made with conventional SD, RHA and FA. Some highlights of the results are as follows:

- The Marshall stability of CMA with RHA and FA filler is 20.32% and 25.99% higher at 9% and 8% OBC than conventional filler mix. The Marshall quotient value of CMA with SD, RHA and FA filler is 1.93 kN/mm, 2.68 kN/mm and 3.01 kN/mm respectively.
- The indirect tensile strength of the mix with RHA and FA is 15.5% and 25% higher than CMA with SD filler. The moisture susceptibility of CMA with SD, RHA and FA is 71.22%, 81.94% and 84.51% respectively, at OBC.
- The rutting resistance of CMA with RHA and FA is 1.66% and 52.16% higher than CMA with SD filler.
- As per the cost analysis, the mixture with FA is 10.11% and 0.16% more cost-effective than CMA with RHA and SD filler.
- The incorporated waste fillers have improved CMA's performance and proved desirable in all aspects. It is found that the FA filler is the most effective, as per the study.

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